1 Planning (15 marks)

Defining the problem (3 marks)

P1 c, d or A is the independent variable and R is the dependent variable or vary c, d or A and measure R. [1]

P2 If c varied then (t and) d or A kept constant, if d varied then (t and) c or A kept constant, if A varied then c or d kept constant. [1]

P3 Keep temperature constant. [1]

Methods of data collection (5 marks)

M1 Circuit diagram to measure resistance. [1]

M2 Use micrometer screw gauge to measure d or t. (Allow digital or vernier callipers) [1]

M3 Measure c with a ruler/metre rule. [1]

M4 Method of making contact with the strip e.g. use electrodes of at least same dimension as c or d or t or conducting paint methods. Do not allow crocodile clips, unless it is clear that the whole area of the end of the strip is covered. [1]

M5 Method to determine resistance. [1]

Method of analysis (2 marks)

A1 Plot a graph of R against c, $1/d$ or $1/A$ depending on orientation. Other alternatives possible, e.g. R against $1/c$ depending on orientation [1]

A2 Must be consistent with A1: $\rho = A \times$ gradient or $t \times$ gradient/c [1]
Other alternatives possible, e.g. $\rho = d \times$ gradient/t

Safety considerations (1 mark)

S1 Reference sharp edges or cutting metals, e.g. wear gloves. [1]

Additional detail (4 marks)

D1/2/3/4 Relevant points might include [4]

1. Insulate aluminium strip
2. Take many readings of t or d and average
3. Use a protective resistor/circuit designed to reduce current
4. Rearrange equation to determine graph using c, d and t or A
5. Determine typical resistance of aluminium strip
6. Likely meter range of ammeter/voltmeter/ohmmeter
7. Detail on cutting strip e.g. mark using set square

Do not allow vague computer methods.

[Total: 15]

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